

Mechanical Brain Hacking : A VR Cybernetics Simulator for Body Editing

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Figure 1: A Player editing her brain and body system. (a) Appearance of the experience. (b) VR world. (c) First person view. (d) Avatar with a third arm

ABSTRACT

Avatar technology has attracted attention, but research in the field of non-humanoid avatars has mainly focused on evaluating the operability and embodiment of new body systems. In addition to these points, we believe that users' design of their own avatar structure and body motion mapping is also important to improve the user experience. We present Mechanical Brain Hacking, a VR cybernetics simulator that allows for easy modification of both the mapping and the structure of the avatar. Participants take on the role of a damaged robot with limited mobility except for the right hand. They use a surgical machine linked to their avatar's hand movements to access their own brain and repair its circuits to regain control of their limbs. Haptic and electrical stimulation feedback, along with visual and auditory feedback through a VR headset, provide a fully immersive experience. Participants can even gain an extra limb by attaching additional limb parts. Through the experience of changing motion mapping and body structure in real time, this project offers a unique perspective on the concept of an "editable body", a body that is edited by oneself and in real time.

CCS CONCEPTS

• **Software and its engineering** → **Interactive games**; • **Human-centered computing** → *Virtual reality*.

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SIGGRAPH '23 Immersive Pavilion, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0151-1/23/08.

<https://doi.org/10.1145/3588027.3595595>

KEYWORDS

Virtual Reality, Human Augmentation, Body Scheme Alternation

ACM Reference Format:

Shuto Takashita, Naoki Tanaka, and Taiga Suzuki. 2023. Mechanical Brain Hacking : A VR Cybernetics Simulator for Body Editing. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '23 Immersive Pavilion)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588027.3595595>

1 INTRODUCTION

In recent years, avatars that enable experiences and abilities beyond those of the natural body have been designed in various fields. For example, in VR games and VRSNS, players use various avatars to gain new experiences and self-expression. These avatars include not only those with a normal human body shapes, but also those with tails, wings, cat ears, and extra arms. The use of such non-humanoid avatars can greatly expand the possibilities of experiences and abilities. In the field of CG animation, for the use of various characters, animators sometimes animate non-humanoid characters such as animals based on human movements, as in the case of Motion Puppetry[Seol et al. 2013]. Also, in the field of human augmentation, research is being done to give humans new abilities by manipulating extended body structures, such as Metalimb[Sasaki et al. 2017]. In this field, avatars exist not only in VR but also in real-life spaces. In non-humanoid avatars, the structure and manipulation method (motion mapping method) of the avatar is critical to its usability.

In the field of non-humanoid avatar research, the current focus has been mainly on the operability of avatars, motion mapping method, embodiment potential, and cognitive changes[Won et al. 2015]. However, in addition to these points, we believe that users

themselves should be able to design their own avatar structure and body motion mapping to improve the user experience. Depending on the experience scenario and task, the optimal avatar structure and motion mapping would vary. They influence the ability to provide the desired experience and ease of use when the user operates the avatar. Allowing users to customize these structure and mapping themselves is therefore an important approach to improving the overall user experience. In addition to allowing users to customize their avatars, real-time confirmation of changes made to the avatars in the VR space is also important. With the ability to make changes and instantly check them, users can achieve a more personalized experience that matches their preferences.

Our project, "Mechanical Brain Hacking," offers participants an experience where they can modify the structure and motion mapping of their own robot avatar in real-time. Taking on the role of a damaged robot that can only move its right hand, they can repair or modify the control of their other limbs by accessing their own brain circuits through a surgical machine linked to their avatar's hand movements. They can attach additional arm parts to their torso, resulting in a customizable body structure. By altering the motion mapping and the body structure in real-time, participants can gain a sense of "editable body", a body that is edited by oneself and in real time. This sense is important in preparing for a future where human augmentation and avatar technologies allow individuals to customize their body structure and operations in real-time, much like adding and removing an accessory.

2 SYSTEM DESIGN

In Mechanical Brain Hacking, participants become a robot with a damaged body (Fig. 1b) and edit their motion mapping by modifying the circuits in their own brain (Fig. 1c) using a surgical machine (Fig. 1b) located on the occipital side of the avatar. We used a robotic avatar rather than a fleshbody avatar to exploit the Proteus effect and make participants feel comfortable modifying their brain and body. Participants can edit brain circuits by remotely controlling the surgical machine while looking at the display in front of the avatar (Fig. 1c). The arms of the surgical machine operate correspondingly to the arms of the avatar. The circuits in the brain consist of a "Consciousness Module" in the center and motion input terminals to each limb in the periphery. The module is equipped with four terminals, each of which outputs the motion of the participant's limbs in reality. (Upper right: right arm, upper left: left hand, lower right: right leg, lower left: left leg) The four motion input terminals also correspond to the motion input to each limb of the avatar in the same order. For example, when the participant moves their real left arm in the state shown in Fig. 2a, the avatar's right arm moves. Similarly, if the participant moves their real right arm in the state shown in Fig. 2b, the avatar's right arm and left leg will move.

Participants can also add a new third arm (Fig. 1d) to their body. By picking up a third arm from the floor of the VR room and attaching it to the left shoulder, a new motion input terminal is added to the brain. By connecting the terminal of one of the Consciousness modules to this terminal, the third arm will move according to the movements of the corresponding limb.

The experience also includes two feedback systems - haptic and electrical stimulation-in addition to a VR headset. Haptic feedback is presented to the back of the head using 4×5 matrix of vibrator

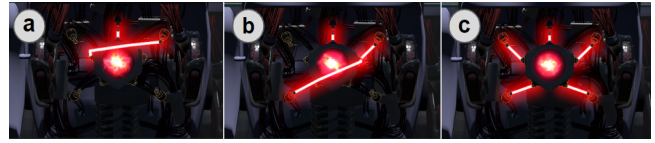


Figure 2: States of avatar's brain which affects their body system. (a) Real left arm moves avatar's right arm. (b) Real right arm moves avatar's right arm and left leg. (c) Avatar movement is completely synchronized with the player.

to simulate the sensation of brain invasion. Electrical stimulation to the limbs simulates "electrical conduction" as the robot gains control of its limbs as the brain circuits are rewired.

3 GAME SCENARIO

"You're a damaged robot with one barely functional arm. To avoid being scrapped, you've decided to crack open your own brain and repair yourself in this basement."

At the beginning, the player's brain circuits is connected as shown in Fig. 2a, and the experience begins with the avatar's left arm being controlled by the player's right arm. The player has two goals: 1) to match the movements of avatar limbs with the player's real limbs, and 2) to add and control a third arm. The player uses the surgical machine's hands to edit brain circuits and restore limbs movement. Along the way, the player may experience a malfunctioning body system due to incorrect circuits connections. By completing the brain circuits (Fig. 2c), the player can match the movements of the avatar to the movements of their real body. Next, the player picks up an arm from the floor of the VR room and attaches it to themselves. They can then edit the brain circuits again and map the desired movements to the third arm. When the player finds their favorite mapping, the experience is complete.

"Congratulations on escaping from the fate of destruction! Please leave the basement and enjoy your new life."

4 DISCUSSION AND FUTURE WORK

In this experience, we have attempted to enable real-time body design by the user. By the end of the experience, we hope to have presented participants with a sense of "editable body" and enabled them to prepare for potential changes to their body system due to future technological advances.

In the future, we will continue to explore how this experience affects participants' body senses and investigate improved real-time design methods for body structure and mapping in VR space.

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